

Saitej Polavarapu

Senior Data Engineer

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SUMMARY

Senior Data Engineer with 5+ years delivering mission-critical data infrastructure at **Amazon** and **Citi Bank**, building large-scale **ETL pipelines**, real-time streaming systems and **LLM-powered AI solutions** across **AWS** and **Azure** using **Python**, **PySpark**, **Apache Airflow**, **Apache Kafka** and **dbt**. Proven expertise in **Amazon Bedrock**, **RAG architectures** and production-grade ML pipelines, delivering scalable and secure data platforms that power AI-driven decision making at enterprise scale. Trusted technical leader with deep **BCBS 239** and **GDPR**-compliant governance expertise across regulated financial environments, consistently shipping high-performing data solutions that drive measurable business impact.

EDUCATION

Master of Science in Computer Science, **University of Dayton**, Ohio, USA

December 2022

Bachelor of Engineering in Electronics and Communication Engineering, **Gitam University**, Andhra Pradesh, India

April 2021

WORK EXPERIENCE

Amazon, CA | Senior Data Engineer

Jan 2024 – Present

- Architected fault-tolerant **ETL frameworks** and data pipelines on **AWS (EMR, Glue, S3, Redshift)** using **PySpark** and **Apache Kafka Streaming**, ingesting 10+ TB daily from 15+ sources with built-in quality checks, reducing pipeline failure rate by 40% and cutting processing latency by 35% across 5+ cross-functional teams.
- Designed and deployed **ML pipelines** on **AWS SageMaker** integrating **Amazon Bedrock** for RAG-based retrieval and LLM-powered data enrichment, leveraging **MLflow** for experiment tracking and model versioning to accelerate delivery for anomaly detection and forecasting use cases.
- Led cross-functional data engineering initiatives across 3+ teams, driving **SQL** query optimization and **Amazon Redshift** performance tuning that reduced cloud infrastructure costs by 30% while mentoring junior engineers on data warehousing best practices and schema design.
- Developed automated event-driven workflows using **Python**, **AWS Lambda**, **Step Functions** and **Amazon EventBridge**, integrating **CI/CD** through **GitHub** and **AWS CDK**, reducing manual deployment efforts by 60% with monitoring through **Amazon CloudWatch**.
- Engineered production data pipelines using **Apache Airflow** and **dbt** for SQL/Python transformations on **AWS (S3, PostgreSQL, Serverless ETL)**, enforcing schema validation and data quality checks across 20+ real-world sources, reducing data inconsistencies by 45%.
- Built a scalable **data lake architecture** on **AWS (S3, Athena, EMR)** using **Hadoop** for distributed batch processing, exposing governed datasets through **REST APIs** and managing infrastructure provisioning through **Terraform** for consistent deployments.
- Implemented secure data migration pipelines using **AWS DMS** and **Amazon RDS** to unify data from **DynamoDB** and **PostgreSQL** sources, enforcing **IAM** roles, encryption policies and audit trails ensuring full compliance with enterprise data governance standards.
- Optimized enterprise data models across **Amazon Aurora** and **Amazon RDS**, applying query optimization and performance tuning that cut execution time by 45%, leveraging **AWS Glue Data Catalog** and **Lake Formation** for metadata management and lineage visibility.
- Deployed real-time streaming pipelines using **Amazon Kinesis** to process high-velocity event streams, transforming data into vector embeddings stored in **pgvector** on **Amazon RDS**, enabling intelligent similarity search for **LLM-powered** applications at low latency.

Citi Bank, NY | Data Engineer

April 2022 – Jan 2024

- Engineered scalable **ETL pipelines** using **Azure Data Factory** and **Azure Data Lake Storage Gen2** to ingest high-volume financial transaction data from 10+ source systems, implementing **CDC pipelines** for real-time change tracking and incremental loads, improving ingestion efficiency by 45% across trade and risk data domains.
- Optimized large-scale data transformation workflows on **Azure Databricks** using **PySpark** and **Delta Lake**, enforcing schema evolution, ACID transactions and data quality standards across financial datasets to ensure consistent and auditable data delivery for regulatory reporting and downstream analytics.

- Maintained and enhanced **SQL Server** databases and **SSIS** packages for banking data ingestion and transformation, implementing snowflake schema methodologies and writing optimized **T-SQL** stored procedures and views to eliminate data discrepancies and ensure integrity across enterprise financial reporting systems.
- Processed high-velocity financial transaction streams using **Azure Event Hubs** and **Azure Synapse Analytics** pipelines, building **Python**-based data validation and reconciliation frameworks to ensure consistent and error-free data delivery across trade and risk banking workflows.
- Enforced enterprise data governance standards using **Azure Purview** for end-to-end metadata management and data lineage tracking, implementing access policies and audit trails aligned with **BCBS 239** risk data aggregation principles and **GDPR** compliance requirements across Citi's financial data infrastructure.
- Streamlined **Azure DevOps** CI/CD pipeline deployments for financial data workflows using **Terraform** Infrastructure as Code and **Git** version control, reducing release cycle time by 35% and eliminating manual deployment errors across regulated banking environments.
- Secured sensitive financial data across **Azure SQL Database** pipelines by implementing **IAM** and **RBAC** access policies and managing secrets through **Azure Key Vault**, ensuring end-to-end encryption and full compliance with Citi's internal security standards and external regulatory audit requirements.
- Configured proactive monitoring and alerting pipelines using **Azure Monitor** and **Log Analytics** to detect anomalies and pipeline failures, applying **Spark optimization** techniques including partitioning, caching and cluster tuning to reduce infrastructure costs by 30% and maintain SLA adherence.

Mu Sigma, India | Data Analyst

Jan 2021 – Aug 2021

- Mined large-scale customer transaction datasets using **Python** and **SQL** to uncover purchasing patterns and trends, delivering data-driven insights that directly supported business decision making across key revenue segments.
- Built interactive dashboards in **Tableau** to visualize critical KPIs including customer retention rate, churn and revenue trends, enabling business stakeholders to monitor performance and make informed strategic decisions.
- Processed and cleansed 500K+ records using **Python** data preprocessing techniques to ensure data accuracy and consistency, eliminating data quality issues before downstream analysis and reporting.
- Partnered with cross-functional teams to translate raw data into actionable business insights, contributing to initiatives that improved sales conversion rates by 12% across key customer segments.
- Automated repetitive data reporting workflows using **Python** scripting, reducing manual reporting effort by 30% and freeing up analyst bandwidth for higher value analytical work.

TECHNICAL SKILLS

Programming Languages: Python, SQL, T-SQL, Scala, Unix Shell Scripting

Big Data & Processing: PySpark, Apache Spark, Hadoop, Delta Lake, EMR

Cloud Platforms: AWS (S3, Redshift, Glue, Lambda, DMS, RDS, Aurora, Athena, EventBridge, Lake Formation, CDK), Azure (Data Factory, Databricks, Synapse Analytics, ADLS Gen2, Event Hubs, Purview, Key Vault, Azure Monitor, Azure DevOps)

ETL & Pipeline Tools: Apache Airflow, dbt, SSIS, Informatica PowerCenter

Streaming Technologies: Apache Kafka, Amazon Kinesis, Azure Event Hubs, Apache Spark Streaming

Databases & Warehousing: SQL Server, PostgreSQL, MongoDB, DynamoDB, Amazon Aurora, Snowflake, Teradata

AI/ML & Libraries: MLflow, Lang Chain, Amazon SageMaker, Amazon Bedrock, RAG Pipelines, Vector Databases, Pandas, NumPy, Scikit-learn

DevOps & Infrastructure: Git, Docker, Kubernetes, Terraform, AWS CDK, CI/CD

Data Governance & Security: IAM/RBAC, Azure Purview, AWS Lake Formation, GDPR, BCBS 239, Encryption

Data Visualization: Tableau, Power BI, Quick Sight, Looker

Methodologies: Agile, SDLC

CERTIFICATIONS

- AWS Certified Machine Learning Specialty
- AWS Certified Data Engineer Associate
- AWS Certified Big Data Specialty